Database System Engineering and Distributed Backend Development

Course Code: 25CS1302E

Practise-2

2520030078

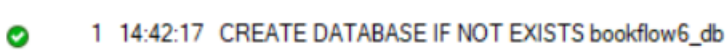
M.Apoorva

Section – 3

Create the database:

```
CREATE DATABASE IF NOT EXISTS bookflow6_db;
```

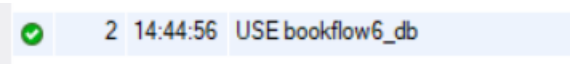
Output:



Use database:

```
USE bookflow6_db;
```

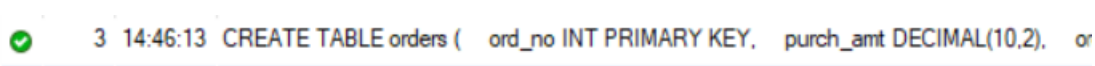
Output:



Create table:

```
CREATE TABLE orders (
    ord_no INT PRIMARY KEY,
    purch_amt DECIMAL(10,2),
    ord_date DATE,
    customer_id INT,
    salesman_id INT
);
```

Output:



Insert data:

```
INSERT INTO orders VALUES
(70001,150.50,'2012-10-05',3005,5002),
(70009,270.65,'2012-09-10',3001,5005),
(70002,65.26,'2012-10-05',3002,5001),
(70004,110.50,'2012-08-17',3009,5003),
(70007,948.50,'2012-09-10',3005,5002),
(70005,2400.60,'2012-07-27',3007,5001),
(70008,5760.00,'2012-09-10',3002,5001),
(70010,1983.43,'2012-10-10',3004,5006),
(70003,2480.40,'2012-10-10',3009,5003),
(70012,250.45,'2012-06-27',3008,5002),
(70011,75.29,'2012-08-17',3003,5007),
(70013,3045.60,'2012-04-25',3002,5001);
SELECT * FROM orders;
```

Output:

ord_no	purch_amt	ord_date	customer_id	salesman_id
70001	150.50	2012-10-05	3005	5002
70002	65.26	2012-10-05	3002	5001
70003	2480.40	2012-10-10	3009	5003
70004	110.50	2012-08-17	3009	5003
70005	2400.60	2012-07-27	3007	5001
70007	948.50	2012-09-10	3005	5002
70008	5760.00	2012-09-10	3002	5001
70009	270.65	2012-09-10	3001	5005
70010	1983.43	2012-10-10	3004	5006
70011	75.29	2012-08-17	3003	5007
70012	250.45	2012-06-27	3008	5002
70013	3045.60	2012-04-25	3002	5001
NULL	NULL	NULL	NULL	NULL

Where Clause:

```
SELECT * FROM orders;
SELECT *
FROM orders
WHERE purch_amt > 2000;
```

Output:

ord_no	purch_amt	ord_date	customer_id	salesman_id
70003	2480.40	2012-10-10	3009	5003
70005	2400.60	2012-07-27	3007	5001
70008	5760.00	2012-09-10	3002	5001
70013	3045.60	2012-04-25	3002	5001
NULL	NULL	NULL	NULL	NULL

Where Clause:

```
SELECT *
FROM orders
WHERE salesman_id=5001;
```

Output:

ord_no	purch_amt	ord_date	customer_id	salesman_id
70002	65.26	2012-10-05	3002	5001
70005	2400.60	2012-07-27	3007	5001
70008	5760.00	2012-09-10	3002	5001
70013	3045.60	2012-04-25	3002	5001
NULL	NULL	NULL	NULL	NULL

Order By:

```
SELECT *
FROM orders
ORDER BY purch_amt DESC;
```

Output:

ord_no	purch_amt	ord_date	customer_id	salesman_id
70008	5760.00	2012-09-10	3002	5001
70013	3045.60	2012-04-25	3002	5001
70003	2480.40	2012-10-10	3009	5003
70005	2400.60	2012-07-27	3007	5001
70010	1983.43	2012-10-10	3004	5006
70007	948.50	2012-09-10	3005	5002
70009	270.65	2012-09-10	3001	5005
70012	250.45	2012-06-27	3008	5002
70001	150.50	2012-10-05	3005	5002
70004	110.50	2012-08-17	3009	5003
70011	75.29	2012-08-17	3003	5007
70002	65.26	2012-10-05	3002	5001
NULL	NULL	NULL	NULL	NULL

Order By

```
SELECT *
FROM orders
ORDER BY ord_date;
```

Output:

ord_no	purch_amt	ord_date	customer_id	salesman_id
70013	3045.60	2012-04-25	3002	5001
70012	250.45	2012-06-27	3008	5002
70005	2400.60	2012-07-27	3007	5001
70004	110.50	2012-08-17	3009	5003
70011	75.29	2012-08-17	3003	5007
70007	948.50	2012-09-10	3005	5002
70008	5760.00	2012-09-10	3002	5001
70009	270.65	2012-09-10	3001	5005
70001	150.50	2012-10-05	3005	5002
70002	65.26	2012-10-05	3002	5001
70003	2480.40	2012-10-10	3009	5003
70010	1983.43	2012-10-10	3004	5006
NULL	NULL	NULL	NULL	NULL

Sum()

```
SELECT SUM(purch_amt) AS total_revenue
FROM orders;
```

Output:

total_revenue
17541.18

Avg()

```
SELECT AVG(purch_amt) AS average_order
FROM orders;
```

Output:

average_order
1461.765000

max():

```
SELECT MAX(purch_amt) AS highest_order
FROM orders;
```

Output:

highest_order
5760.00

Count():

```
SELECT COUNT(*) AS total_orders
FROM orders;
```

Output:

	total_orders
▶	12

Group By:

```
SELECT salesman_id,
       SUM(purch_amt) AS total_sales
FROM orders
GROUP BY salesman_id;
```

Output:

	salesman_id	total_sales
▶	5002	1349.45
	5001	11271.46
	5003	2590.90
	5005	270.65
	5006	1983.43
	5007	75.29

Group By:

```
SELECT customer_id,
       SUM(purch_amt) AS total_purchase
FROM orders
GROUP BY customer_id;
```

Output:

	customer_id	total_purchase
▶	3005	1099.00
	3002	8870.86
	3009	2590.90
	3007	2400.60
	3001	270.65
	3004	1983.43
	3003	75.29
	3008	250.45

Group By + Max():

```
SELECT customer_id,
       MAX(purch_amt) AS highest_purchase
FROM orders
GROUP BY customer_id;
```

Output:

customer_id	highest_purchase
3005	948.50
3002	5760.00
3009	2480.40
3007	2400.60
3001	270.65
3004	1983.43
3003	75.29
3008	250.45

Group By + Having

```
SELECT salesman_id,
       SUM(purch_amt) AS total_sales
FROM orders
GROUP BY salesman_id
HAVING SUM(purch_amt) > 3000;
```

Output:

salesman_id	total_sales
5001	11271.46

Group By + Having:

```
SELECT customer_id,
       COUNT(*) AS total_orders
FROM orders
GROUP BY customer_id
HAVING COUNT(*) > 1;
```

Output:

customer_id	total_orders
3005	2
3002	3
3009	2

Group By + Having + Order By:

```
SELECT customer_id,
       SUM(purch_amt) AS total_purchase
FROM orders
GROUP BY customer_id
HAVING SUM(purch_amt) > 1000
ORDER BY total_purchase DESC;
```

Output:

customer_id	total_purchase
3002	8870.86
3009	2590.90
3007	2400.60
3004	1983.43
3005	1099.00

Between + Having:

```
SELECT customer_id,
       MAX(purch_amt) AS max_purchase
FROM orders
GROUP BY customer_id
HAVING MAX(purch_amt) BETWEEN 2000 AND 6000;
```

Output:

customer_id	max_purchase
3002	5760.00
3009	2480.40
3007	2400.60

Count + Having:

```
SELECT salesman_id,
       COUNT(*) AS total_orders
FROM orders
GROUP BY salesman_id
HAVING COUNT(*) >= 2;
```

Output;

salesman_id	total_orders
5002	3
5001	4
5003	2

Max + Group By + Having:

```
SELECT salesman_id,
       COUNT(*) AS total_orders
FROM orders
GROUP BY salesman_id
HAVING COUNT(*) >= 2;
```

Output:

	salesman_id	total_orders
▶	5002	3
	5001	4
	5003	2